Assigment6

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1 Question_0 – Website Results

There is a huge change, that I went to a complete different direction in this segment sorry from ahead.

(a) Which algorithm is used?

The site runs the following simple loop:

Algorithm 1 Iterative Maximum-Weight Matching for Course Allocation

while some student still needs credits **do**
  Publish how many seats are left in every course.
  Build a bipartite graph:
  - Left side: one node per student who still needs credits.
  - Right side: one node per free seat.
  - Edge weight: that student's valuation of the course.

  Match – run a maximum-weight matching on the graph. Each student wins (at most) one course.
  Allocate those seats and delete them from the pool.
end while

This is often called *Iterative Maximum-Weight Matching*. You can also view it as a "parallel" greedy round-robin: everyone gets exactly one course per round, but the choice is optimised for the whole group at once.

(b) Why is the result EF1?

Let B_i be the final bundle of courses that student i gets.

- Look at the first round where i likes a course given to j more than the one he got in that same round.
- In all earlier rounds i already picked (by the matching) a course that he liked at least as much as any other seat still available.
- If we delete *that one course* from j's bundle, i no longer prefers j's bundle to his own.

So for every pair of students i, j, i may envy j at most by a single course—exactly the definition of EF1 (envy-free up to one item).

Bottom line: the platform uses an iterative max-weight matching routine, and the allocation it outputs is always EF1.

2 Question_2

Setting. Let G be the set of items, $N = \{1, \ldots, n\}$ the players, $v_i(g)$ the *true* value of player i for item g and Let b_g be the bid that $Player_1$ declares for $item(g)$. Under *Equal-Center* each item goes to the highest declared value and the winner pays his own declaration; afterwards every player receives the same share $\frac{M}{n}$ where M is the total money collected.

Key observation. Fix a item g and write

$$h_g \;=\; \max_{j \neq 1} v_j(g) \quad \text{(highest rival value)}.$$

If $Player_1$ bid satisfies this condition $b_g > h_g$, $Player_1$ will get the item g and his utility would be

$$u_g^{\text{win}} \;=\; v_1(g) - b_g \;+\; \frac{b_g - h_g}{n},$$

because the payment increases the pool by $(b_g - h_g)$ and he gets $\frac{1}{n}$ of that back. If instead he loses, his utility is

$$u_g^{\text{lose}} = 0.$$

Choosing the minimal winning bid $b_g = h_g + \varepsilon$ (with infinitesimal ε) makes the gain (over losing)

$$u_g^{\text{win}} - u_g^{\text{lose}} \approx v_1(g) - h_g.$$

Hence winning is worthwhile *iff* $v_1(g) > h_g$.

Algorithm 2 Optimal Deviation for $Player_1$ (full information)

Require: $Player_1$'s values $v_1(g)$ and all rivals' values $v_j(g)$, $j \neq 1$

for each item $g \in G$ **do**
  $h_g \leftarrow \max_{j \neq 1} v_j(g)$
  if $v_1(g) > h_g$ **then**
    $bid_g \leftarrow h_g + \varepsilon$ ▷ minimal winning bid
  else
    $bid_g \leftarrow 0$ ▷ or any value $\leq h_g$ (intuitively irreverent since we wouldn't bid more than the value)
  end if
end for
return the bid vector $\big(bid_g\big)_{g \in G}$

Optimal strategy. This rule maximises $Player_1$'s total utility because for each item it implements exactly the cases where the marginal gain $v_1(g) - h_g$ is positive. □

(b) Why no "safe" manipulation exists without any information

Assume now that $Player_1$ *does not* know the rivals values and suppose he adopts a fixed mis-reporting rule that differs from truth-telling. We show the rule can hurt him in some feasible profile, so it is not risk-free.

- Take a single item g and let v_1 be $Player_1$'s true value.
- Construct two rival profiles that demonstrate the possibility of the manipulation not working, by both increase and decreasing the value of item g:
  (i) **Profile A:** the highest rival value is $h_A = v_1 - \epsilon$ (just below v_1).
  (ii) **Profile B:** the highest rival value is $h_B = v_1 + \epsilon$ (just above v_1),

with the same small $\epsilon > 0$.

- If $Player_1$ *tells the truth*, he
  - wins and gains a positive surplus in Profile A;
  - loses and pays nothing in Profile B,

  so his payoff is non-negative in both cases.

- Any non-truthful declaration b must eithis overshoot v_1 or undershoot it:
  - If $b < v_1$, then in Profile A he *loses* the item, dropping his utility below truthful bidding.
  - If $b > v_1$, then in Profile B he *wins* but pays more than the item is worth to him, again lowering his utility.

Because $Player_1$ cannot know which profile is real, *any* manipulation strictly decreases his payoff in at least one possible world. Therefore no manipulation is safe. □